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ORIGINAL PAPER

Simulating flood situations in urban hydrology using a diffusion model in complex networks

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Abstract
Poorly planned urbanization combined with climate change increases the annual likelihood of flooding in low-lying areas with inadequate storm drainage systems. Therefore, the development of models to predict flood situations in cities is crucial because floods pose significant risks to human lives and infrastructure. This paper utilizes the diffusion model in complex networks to simulate flood scenarios in the specific context of urban hydrology. By adapting the model, this study describes how rainwater spreads through interconnected nodes within the network. These nodes represent different areas of a city, such as buildings, roads, and water bodies. Various properties, including elevation, surface roughness, and permeability, were assigned to nodes that influence the flow of water. The edges between nodes represent pathways for water movement. The model considers factors such as rainfall intensity, network topology, properties, and existing flood mitigation measures. The proposed model was evaluated in two experiments. The first experiment aimed to identify the critical points in the network, whereas the second experiment examined how the number and distribution of storm drains affected flooding. As a case study to test the model, it was tested in a flood-prone section of Guadalajara, Mexico. The experimental results demonstrate that our model is flexible and simple, allowing us to draw interesting conclusions from several flooding scenarios.

Keywords Agent-based modeling · Flood risk prediction · Dynamic diffusion system · Complex network

1 Introduction

Urban expansion into undeveloped land significantly alters natural landscapes, affecting water flow and increasing surface runoff, which can overwhelm stormwater systems and cause flooding. Urban floods lead to problems that arise every year, including pollution, health hazards, structural damage, and loss of life (Poppema 2014). Furthermore, flood risk is escalating in vulnerable regions, with climate change expected to worsen these impacts (Jenkins et al. 2017). Predictive models for floods and similar phenomena are critical, as they identify risk-prone areas to help mitigate these events (Amankwah-Amoah et al. 2021; Liu et al. 2020).

Recent advancements in computational methods for flood prediction stem from increased access to high-performance computing and innovative modeling techniques. Generally, these methods fall into three categories: hydrological models, data assimilation techniques, and machine learning methods.

Hydrological models simulate water flow within urban environments by considering variables such as rainfall intensity, runoff, and infiltration, helping predict flood likelihood and severity (Kauffeldt et al. 2016; Wijayarathne et al. 2020; Icyimpaye et al. 2022). These models often require detailed input data, such as topographic, land cover, and soil data, which can be costly and challenging to gather, especially in urban settings with limited or outdated data.

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Data assimilation techniques integrate real-time rainfall and water level measurements to enhance model accuracy and enable real-time flood forecasting (Wu et al. 2020; Ziliani et al. 2019; Di Mauro et al. 2022). These techniques rely heavily on high-quality observational data, which, when scarce or unavailable, can hinder their effectiveness. Machine learning methods utilize historical data to forecast flood risk, accounting for land use, soil type, and topography (Nguyen et al. 2020; Ding et al. 2020; Sahoo et al. 2021). As with other data-driven approaches, the accuracy of machine learning models is dependent on the quality and volume of data available, with large, high-quality datasets being essential for reliable predictions. Despite the advances in these three methods, two main challenges persist. First, they rely on complex numerical models and algorithms, demanding considerable computational power and time. Second, their complexity can make it difficult to understand how they reach their predictions, posing challenges for interpreting and communicating model outcomes, which are crucial for informed decision-making and generating flood risk maps.

Traditional modeling techniques often struggle to capture highly complex phenomena. In such cases, alternative approaches like agent-based modeling and complex networks offer effective solutions (Wilensky and Rand 2015). Agent-based models (ABMs) emphasize individual agents' types of behavior and interactions with each other and their environment. Each agent operates with unique properties, rules, objectives, and decision-making processes, leading to emergent behavior from these interactions.

A complex network, on the other hand, is a system composed of interconnected components, or nodes, that exhibit complex behavior through their connections (Sayama 2015). Represented graphically with nodes as components and edges as connections, complex networks are ideal for systems where interconnections between elements are crucial. These networks are characterized by a topology that significantly influences system behavior (Macal et al. 2005), with emergent properties arising from the interactions among components. This focus on overall structure allows complex networks to model systems that are difficult to capture with classical techniques. In complex networks, the emphasis is on network structure and topology—connections, dependencies, and network properties like connectivity, clustering, centrality, and modularity—which influence node behavior independently of the individual agents. Although ABMs and complex networks differ in focus, they share similarities: both represent system interactions and structure through nodes or agents. Consequently, concepts and properties from agent-based modeling can often be applied to complex networks and vice-versa. Complex networks have diverse applications across fields such as social sciences, biology, transportation, and medicine (Yang et al. 2020; Messier et al. 2019; Mou et al. 2020; Sun et al. 2020; Sawai 2012; Hofmann et al. 2016; Cherifi et al. 2019; Boers et al. 2019). They can demonstrate dynamic behavior, especially when modeled as dynamical networks where node interactions evolve over time. In these networks, each node's behavior is influenced by its connections, which may change in strength or structure, leading to emergent behavior across the networask.

The diffusion model is a dynamic approach used to describe the spread of information or elements—such as heat, fluids, or particles—through a complex network. In this model, each network node has a value representing its state, which serves as an initial condition reflecting the network's starting configuration. As the diffusion process unfolds, nodes interact with their neighbors, updating their values based on these interactions. Typically, values are updated using a form of averaging or weighted combination of neighboring values, although specific rules may vary by model. This process gradually propagates the initial values throughout the network as nodes influence and are influenced by their neighbors, spreading the initial state over time. The diffusion speed and extent depend on factors like network topology, connection strengths, and the initial conditions themselves. The diffusion model is widely applied across research fields, including sociology, image processing, artificial intelligence, and medicine, to analyze the spread of innovations, ideas, behavioral patterns, and diseases within populations (Wu et al. 2019; Borowski et al. 2020; Ali et al. 2023; Chen et al. 2023; Zhang and Agrawala 2023; Rombach et al. 2022; Zhang et al. 2022; Firdaniza et al. 2021).

In this study, we adapt the diffusion model to simulate flood scenarios within urban hydrology, conceptualizing water spread through a network of interconnected nodes. Here, nodes represent different urban locations, such as buildings, roads, or water bodies, each assigned properties like elevation, surface roughness, and permeability, which influence water flow. Edges between nodes represent potential pathways for water movement. The model incorporates factors such as rainfall intensity, network topology, and flood mitigation measures, enabling it to simulate water diffusion across various urban scales, from small city areas to extensive regional networks. One key advantage of using the diffusion model for flood simulation is its adaptability to networks of varying complexity, while its straightforward structure facilitates easy observation and interpretation of results throughout the diffusion process.

The remainder of this paper is organized as follows: Sect. 2 introduces the basic concepts of complex networks. In Sect. 3, the dynamic diffusion of the network model is reviewed. Section 4 presents the proposed flood-risk model.

In Sect. 5, the experimental results and a validation analysis are presented. Finally, conclusions are drawn in Sect. 6.

2 Complex networks

A complex network is a set of interconnected nodes or entities that are characterized by non-random patterns of connectivity. These networks can be found in a variety of natural, social, and technological systems, such as the Internet, biological organisms, social networks, and transportation networks. Figure 1 shows a typical example of a complex network. In a complex network, each node n_i represents an individual component of the system while the edge $E_{i,j}$ corresponds to the connection between the node n_i and n_j.

In a complex network, the nodes can represent individuals, groups, or physical entities, while the edges or links between the nodes represent the interactions or relationships between them. These interactions can take many forms, such as social ties, physical connections, or information flows. One way to characterize complex networks is through their topology, which refers to the patterns of connections between nodes. Different types of network topologies can exhibit unique properties, such as small-world properties, scale-free properties, or modular structures. These properties can influence how information, resources, or influence spread through the network.

A complex network can be used as a modeling approach to represent a system of many interconnected components, where the interactions between these components give rise to complex behavior. The main characteristics of complex networks for modelling are their emergent properties and topology. Complex networks can exhibit emergent properties that arise from the interactions between individual components of the system. These emergent properties can be difficult to predict using traditional modeling approaches.

Traditional equation-based models, while effective in many contexts, often assume deterministic and linear relationships among system components. This assumption makes it difficult to incorporate stochastic interactions that play a critical role in complex systems. These models may produce overly deterministic responses, thus limiting the possibility of capturing different types of emergent behavior.

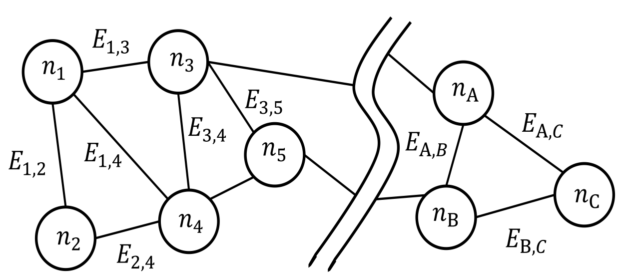


Fig. 1 Example of a complex network

Complex networks, on the other hand, excel in representing systems with dynamic and non-linear interactions between nodes. This flexibility allows for the appearance of emergent properties as individual nodes interact in ways that lead to unpredictable, large-scale patterns within the network.

The arrangement of nodes and edges in the network is known as the network topology. This topology can have a significant impact on the behavior of the system, and different types of topology can give rise to distinct behavioral patterns.

Until this point, we have described static complex networks, where nodes and their connections remain fixed. However, many real-world systems evolve over time. To represent this kind of behavior, it is possible to model complex networks as dynamical systems, where both nodes and connections can change. These are known as dynamical networks.

A dynamical network is a network of interconnected nodes or entities that are characterized by their ability to change and evolve over time. In contrast to static networks, where the topology and connections between nodes remain fixed, dynamical networks are characterized by the presence of dynamic processes that cause the network structure to change and evolve.

A dynamical network operates through a set of processes that cause the structure of the network to change and evolve over time. In general, these processes can include node dynamics and/or edge dynamics. Node dynamics refers to the case that nodes $n_i(k)$ or their state in a dynamical network can change depending on the iteration k while edge dynamics corresponds to the fact that the connections $E_{i,j}(k)$ of the complex network can be altered, added, or removed over time.

Dynamical networks can be modeled using a variety of mathematical and computational techniques, such as differential equations, cellular automata, or agent-based models. These models can capture the dynamics of the network structure over time, as well as the interactions between nodes. One important feature of dynamical networks is their ability to exhibit emergent behavior, where complex patterns or phenomena emerge from the interactions between individual nodes. For example, in a social network, the spread of ideas or types of behavior can give rise to larger-scale phenomena such as social norms or cultural trends.

3 Dynamic diffusion on a network model

The diffusion model is a type of dynamic model that describes the spread of numerical information through a complex network. This model is a generalization of spatial diffusion models in nonregular and nonhomogeneous spatial topologies. Each node n_i is associated with a state c_i that represents a local place where something can be accumulated. Each symmetrical edge $E_{i,j}(\ E_{j,i})$ represents a channel through which something can be transported, one way or another, driven by the gradient of its concentration, as follows:

$$\frac{dc_i}{dt} = \alpha \sum_{j\epsilon N_i} (c_j - c_i) \tag{1}$$

where α represents the diffusion constant and N_i is the set of nodes that are neighbors of node i. $(c_j - c_i)$ is the difference in the concentration between nodes j and i across the edge (i, j). If neighbor n_j has a higher concentration (higher state value) than node n_i, there is an influx from n_j to n_i, causing a positive effect on dc_i/dt. The negative effect on dc_i/dt is produced if neighbor n_j has a lower concentration (a lower state value) than node n_i; then, there is an outflux from n_i to n_j. Therefore, Eq. (1) can be written as follows:

$$\frac{dc}{dt} = \alpha L(c_j - c_i) \tag{2}$$

where L is the Laplacian matrix of the network, which is defined as $L = D - A$, where A is the adjacency matrix of the network and D is the degree matrix of the network. The Eq. (1) can be discretized in time, as shown in Eq. (3) to simulate the dynamics.

$$\begin{aligned} c_i(t+\Delta t) &= c_i(t) + \left[\alpha \sum_{j\epsilon N_i} (c_j(t) - c_i(t))\right]\Delta t \\ &= c_i(t) + \alpha\left[\left(\sum_{j\epsilon N_i} c_j(t)\right) - c_i(t)\ \deg(i)\right]\Delta t \end{aligned} \tag{3}$$

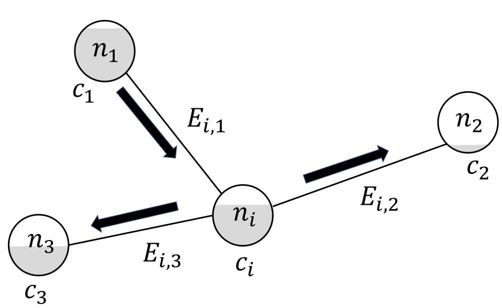


Fig. 2 Diffusion process

The Laplacian representation (Eq. 2) can also be discretized in time as shown in Eq. (4).

$$c(t+\Delta t) = c(t) - \alpha Lc(t)\Delta t = (I - \alpha L\Delta t)\, c(t) \tag{4}$$

where c is the state vector for the entire network and I is the identity matrix. Figure 2 illustrates the diffusion process. In the Fig. 2, node n_i has three neighbors. The level of each node (grey representation inside the node) corresponds to the state value. According to the Eq. (1); the arrows indicate the flow direction of the diffusion process.

The diffusion process can also be applied to directed networks, in which the direction of edges affects the way information spreads.

In this model, nodes in the network are assigned a state variable. The state of each node is then updated over time based on the states of its neighbors in the network. In its operation the model considers four elements: Initialization, Diffusion rule and Network structure.

In the initialization, the initial state of each node is set to a particular value. At each time step k, the state of each node is updated based on the states of its incoming neighbors. The precise update rule is based on differences of neighbor states such as it is shown in Eq. 2. The structure of the directed network can also play an important role in the diffusion process. For example, the directionality of edges can affect the speed and direction of information or disease spread, and the presence of cycles or feedback loops can lead to amplification or damping of the diffusion.

In each iteration, each node shares the percentage of its state with its neighbors in the network. Specifically, the shared value is divided equally and sent along each node's outgoing links. A node with no outgoing links does not share any value; it simply accumulates when its neighbors are provided via inbound links. Figure 3 shows the process of diffusion on a direct network.

The size of each node in Fig. 3 corresponds to the state of the node, where the node's area is proportional to its value. Figure 3a shows a directed network with the initial state of each node. As shown in the Figure, all nodes are initialized with the same state value. The diffusion process was then applied over the network. Figure 3b–d present the diffusion results for iterations 15, 50, and 300, respectively. As can be seen, information tends to accumulate in nodes with no outgoing links.

4 Model description

This section explains the proposed model for flood risk prediction that implements a dynamic diffusion system in a complex network. The diffusion model in complex

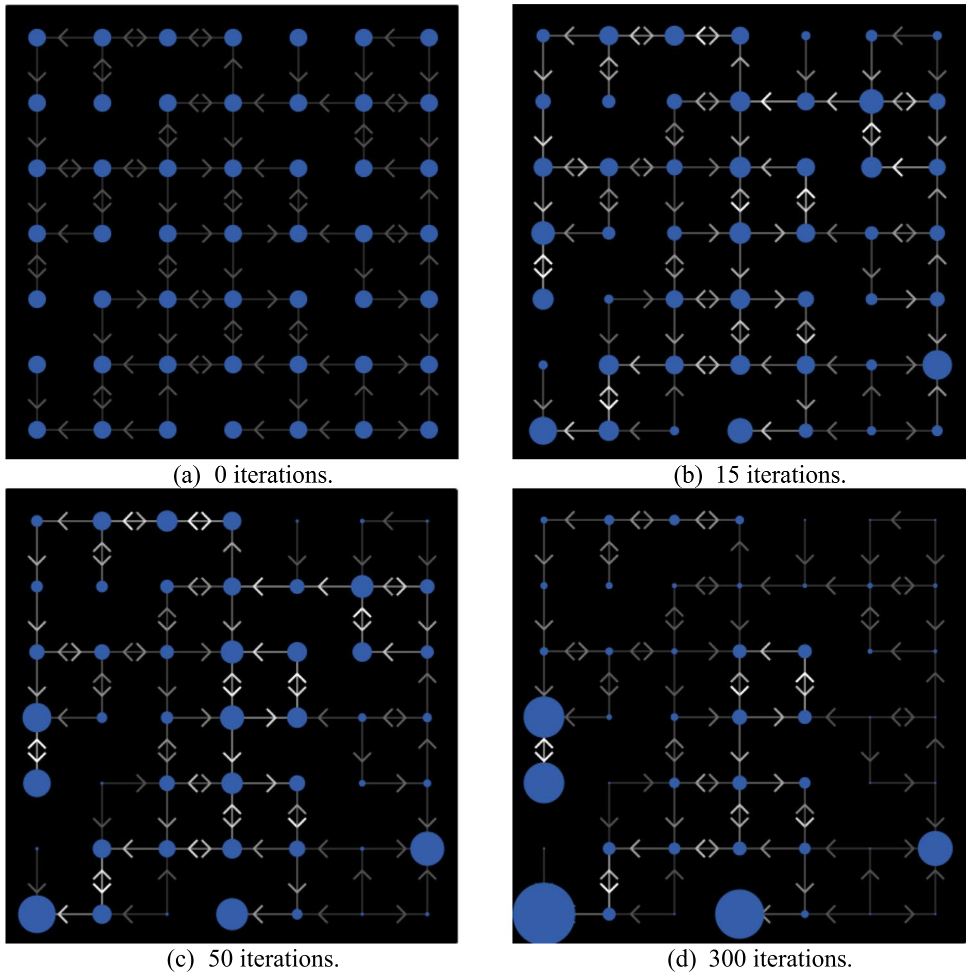


Fig. 3 Diffusion model on a directed network

networks was used to simulate flood situations by adapting it to the specific context of urban hydrology. Under this approach, the spread of rainwater is described by how it diffuses through a network of interconnected nodes. This section is divided into two parts: (A) the model description and (B) the computational procedure.

4.1 Model description

The probability that a certain area of a city will flood depends on several factors, such as rainwater infrastructure, elevation of the area, soil permeability, and garbage. In our approach, the floods in a city are modeled by implementing water accumulation and diffusion rules under a complex network model.

The nodes in the complex network were strategically selected to represent key intersections within the city where flooding is most likely to occur or where water movement is critical. These nodes correspond to intersections of streets in flood-prone areas, chosen based on historical flood data and hydrological studies (Instituto de Planeación y Gestión del Desarrollo del Área Metropolitana de Guadalajara, UNAM 2021), (Instituto de Información Estadística y Geográfica del Estado de Jalisco 2015). Each node represents an intersection or critical point within the urban network, incorporating factors such as elevation, surface characteristics, and the hydrological connectivity of the area. This selection ensures that the model accurately captures the city's vulnerability to flooding.

The nodes of the complex network represent intersections of streets in an area of a specific city. Each node has three parameters: elevation, water quantity, and diffusion rate. The elevation profile of the zone influences the direction in which water diffusion occurs between nodes. The diffusion rate parameter emulates different conditions such, as the number of sewers, quantity of garbage, infrastructure capacity, slope steepness, and area permeability.

The elevation and diffusion rates influence the water accumulation in each node and the diffusion between the nodes.

The proposed model is designed to be adaptable to various urban areas beyond Guadalajara by incorporating geographical and topological data specific to the new region. For applications in other locations, key parameters such as elevation, land use, infrastructure layout, and soil permeability should be configured to reflect the characteristics of the target area. The complex network representing the urban layout can be generated using models like Watts and Strogatz (1998) or similar frameworks, allowing for the replication of realistic connectivity and spatial patterns seen in urban environments.

The following subsections explain the creation of the network, initial parameter values assigned to each node, diffusion process, and behavioral rules of the complex network.

4.1.1 Creation of the network

A network is created to represent a given city area. A set of nodes in the network represents each street intersection. The links between nodes represent the roads through which the water flows. The absolute value of the elevation difference determines the weight of the links between two nodes. Figure 4a shows an example of a city map and Fig. 4b illustrates the equivalent network.

4.1.2 Parameter settings and initialization

In the proposed model, the network is represented by a set of nodes $A^k = \{a_1^k, a_2^k, \ldots, a_N^k\}$, where N is the total number of nodes and k represents each iteration in the computation of the system. Each agent a_i $(i \in 1, \ldots, N)$ has three attributes: elevation, water quantity, and diffusion rate.

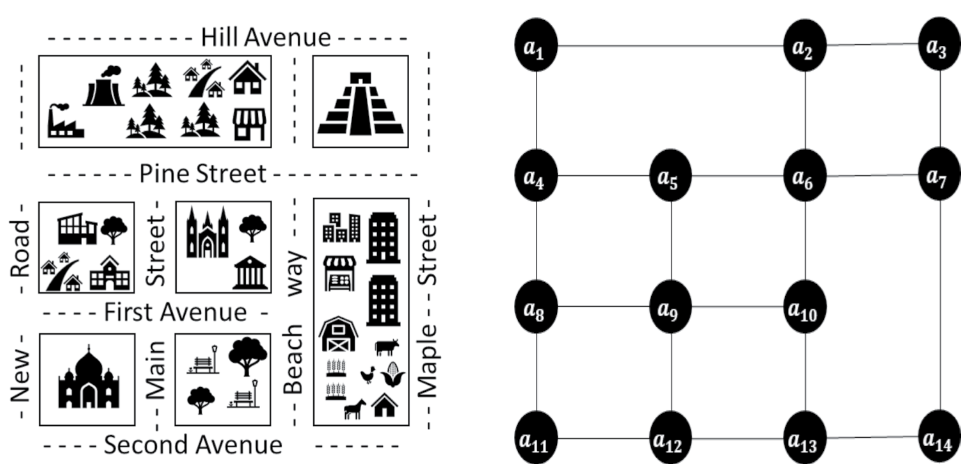
The elevation parameter above the sea was set by considering the actual coordinates of the geographical location of the node. The water quantity parameter is initialized with a random value within a uniform distribution $D[0{,}100]$, representing a relative measure of water content at each node. This value is an abstract metric rather than a specific physical unit, allowing the model to be applied flexibly across various scenarios. In our model, this parameter may change at each iteration k to simulate dynamic changes in water levels. Finally, the diffusion rate parameter was set to a uniform distribution, $D[0{,}1]$. This parameter is important because it considers factors such as the number of sewers, quantity of garbage, infrastructure capacity, slope steepness (absolute value of the elevation difference), and permeability of the area. The diffusion rate parameter may also change at each iteration k.

The model begins by defining the network structure. Therefore, the neighbor nodes of each element a_i were determined using the adjacency matrix of the network. The output direction of diffusion is indicated by the node with the highest elevation among its neighbors. The input directions of diffusion were established for the nodes with the lowest elevation among their neighbors. Figure 5 shows the initial parameter settings of a section of the network, as shown in Fig. 4b, using the proposed model.

The initial representation of the parameter settings is shown in Fig. 5, which indicates that nodes a_9 and a_5 have the same elevation. Hence, the link weight is zero and exhibited bidirectional diffusion. Node a_{10} has a lower elevation; therefore, it has an incoming diffusion influence from node a_9. In its operation, node a_{10} receives a quarter (0.4%) of the water quantity from node a_9. Likewise, a_{10} distributes 0.2% among its neighbors a_6 and a_{13} if the elevation allows it.

4.1.3 Behavioral rules

In our model, the nodes from set A of the network follow specific behavioral patterns defined by rules. These rules allow for correct interactions between nodes and the environment. These simple rules of the model led to interesting patterns related to network topology. This set of rules allows us to emulate the flooding process through a complex network. These rules consider several aspects that add interesting effects to the simulation.



(a) Example of a city map.

(b) City network graph.

Fig. 4 City network graph representation

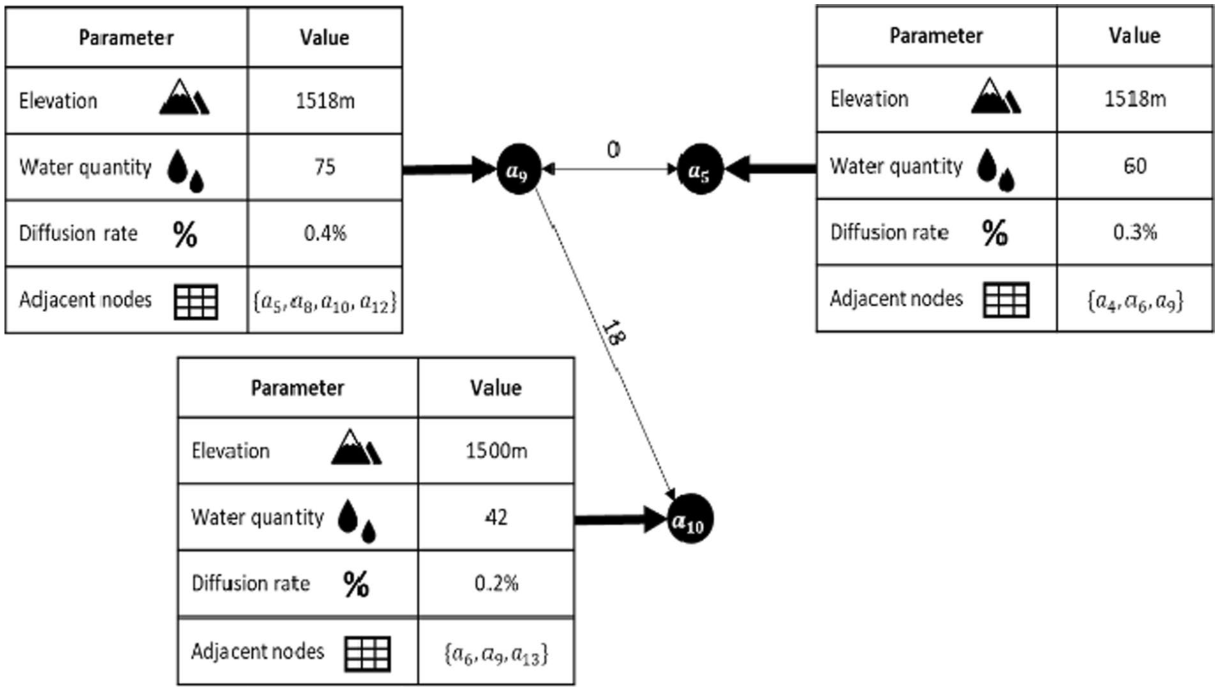


Fig. 5 Initial parameter settings

The model consists of two rules. The first rule establishes the directions and amount of water a node transmits to neighboring nodes by its diffusion rate. This rule is fundamental because it indicates the direction in which water should flow from each node. It also shows the nodes with more influence of inputs or outputs, such as those that are more prone to flooding or less prone to flooding. The main objective of the second rule is to emulate the accumulation of water in a node owing to the decrease in the water diffusion rate. Another objective of this rule is to change the diffusion rate to a higher percentage than the original rate.

4.1.3.1 Rule I– Direction and water diffusion rate The first rule establishes the direction and amount of water a node transmits to neighboring nodes by its diffusion rate. To define which nodes transmit or receive water, it is important to determine the elevation parameter of each node from the network by considering the actual coordinates of the geographical location of the node. Therefore, from the set of nodes A of the network, it is verified whether they have a higher elevation level than their neighbors. In order to illustrate this process, Fig. 4. If it is the case that a_9 has a higher elevation level than a_{10}, it means that the outward diffusion direction will be influenced from a_9 towards a_{10}. If this condition is not satisfied, the inward diffusion influence will be from a_{10} to a_9. The diffusion rate is controlled by the number of sewers, quantity of garbage, infrastructure capacity, slope steepness (absolute value of the elevation difference or weight), and permeability of the area.

The diffusion rate indicates the percentage of water that a node donates to its neighbors with lower elevation levels in each iteration. Specifically, the amount of shared water is divided and sent along the outgoing links from each node to other nodes. If a node has no outgoing links, it does not share any of its water; it simply accumulates any water provided by its neighbors via incoming links.

According to Fig. 4, a_9 has a diffusion rate of 0.4%, and it has two available neighbors, a_5 and a_{10}. It will transmit 0.4% of the water flow between the two. In other words, a_9 sends 0.2% of water to each node. The absolute value of the elevation difference between a_9 and its neighbors indicates a distribution of 0.4%. Assuming that the weight between a_9 and a_{10} is 18 and the weight between a_9 and a_5 is 0, the water quantity from a_9 to a_{10} will be distributed with 2/3 of 0.4% and with 1/3 of 0.4% from a_9 to a_5.

4.1.3.2 Rule II– Water accumulation and diffusion rate change The second rule aims to emulate complex situations, such as the accumulation and release of garbage and the failure of rainwater infrastructure at each node. To simulate this behavior, our model includes a certain probability that the diffusion rate of each node will change by decreasing or increasing its value in a proportion as the diffusion process evolves. This change in diffusion rate simulated the stochastic accumulation or release of water. The probability of a decrease in the diffusion rate is $p_d = 0.3$. The probability of the diffusion rate increase was $p_u = 0.1$. Under such conditions, the operation of the rule is as follows. Assuming a determined iteration k and the selection of a random node a_i, we first produce a uniformly distributed random number r_1. If r_1 was less than 0.3, the diffusion rate decreased by 10%. Then, a new random number r_2 that is uniformly distributed is generated. If r_2 was less than 0.1, the diffusion rate increased by 10%. If probabilistic events do not occur, then the diffusion rate remains unchanged. The operation of this rule did not occur in any iteration k. This is applied for each d_c iterations.

4.2 Computational procedure

The proposed model has been implemented as an iterative approach. The model requires the city's geographical data (coordinates and elevation) of the area to be analyzed, the number of iterations (maxIter), the individual indicator of change in the diffusion rate d_c, and the probabilities p_d and p_u. Algorithm 1 provides an overview of the model. Algorithm 1 consists of four sections: input data, creation of the network and initialization, and rule I and rule II.

The first step in the model was to create a network from the city's imported geographic data. Tools such as OpenStreetMap simplify the export of geographical data. Subsequently, all nodes A from the network and adjacency matrix A_m are obtained to identify the neighbors of each node. Individual parameters of the initial water quantity w_q and diffusion rate d_r were randomly generated for each node. Subsequently, elevation parameters were set for each node. The weight between connections is determined by the absolute value of the elevation difference between a node and its adjacent neighbor node.

The algorithm begins to operate in an iterative loop by applying the first behavioral rule. The first rule starts by evaluating the diffusion directions of the current agent and then identifies the number of nodes it influences to distribute a part of the value of its water quantity among its neighboring nodes.

The second rule is executed after the first rule as long as the individual counter variable d_c allows it. In this section, the value of the diffusion rate of the current agent is affected, generating an increase or decrease in the current value if the probabilities p_d or p_u allow it. Finally, the iterative cycle continues to select from one agent to another until the stopping criterion is equal to the number of system iterations ($maxIter$)

Input data: Geographical data (coordinates, elevation), $d_c, maxIter, p_d, p_u$	
$A, A_m \leftarrow$ networkCreation($Geographical\ data$) $w_q, d_R \leftarrow$ randomParameters(A) $p_d, p_u \leftarrow 0.3$ $d_c \leftarrow 0$	Creation Of the Network and Initialization
while $Iter \leq maxIter$ **do** **for each** A_i **do**	

if elevation of $A_i \geq$ elevation of A_m **do**
$A_m, w_q \leftarrow$ waterDiffusion(A_i, w_q, d_r)
end if
	Rule I

if $d_c \neq 0$ **do**
if $p_d \mid\mid p_u > rand[0,1]$ **do**
$A_i, d_c, d_r \leftarrow diffusionChange(A_i, d_r, p_d, p_u)$
end if
end if
	Rule II

end for **end while**	

Algorithm 1 Pseudo-code of the proposed model

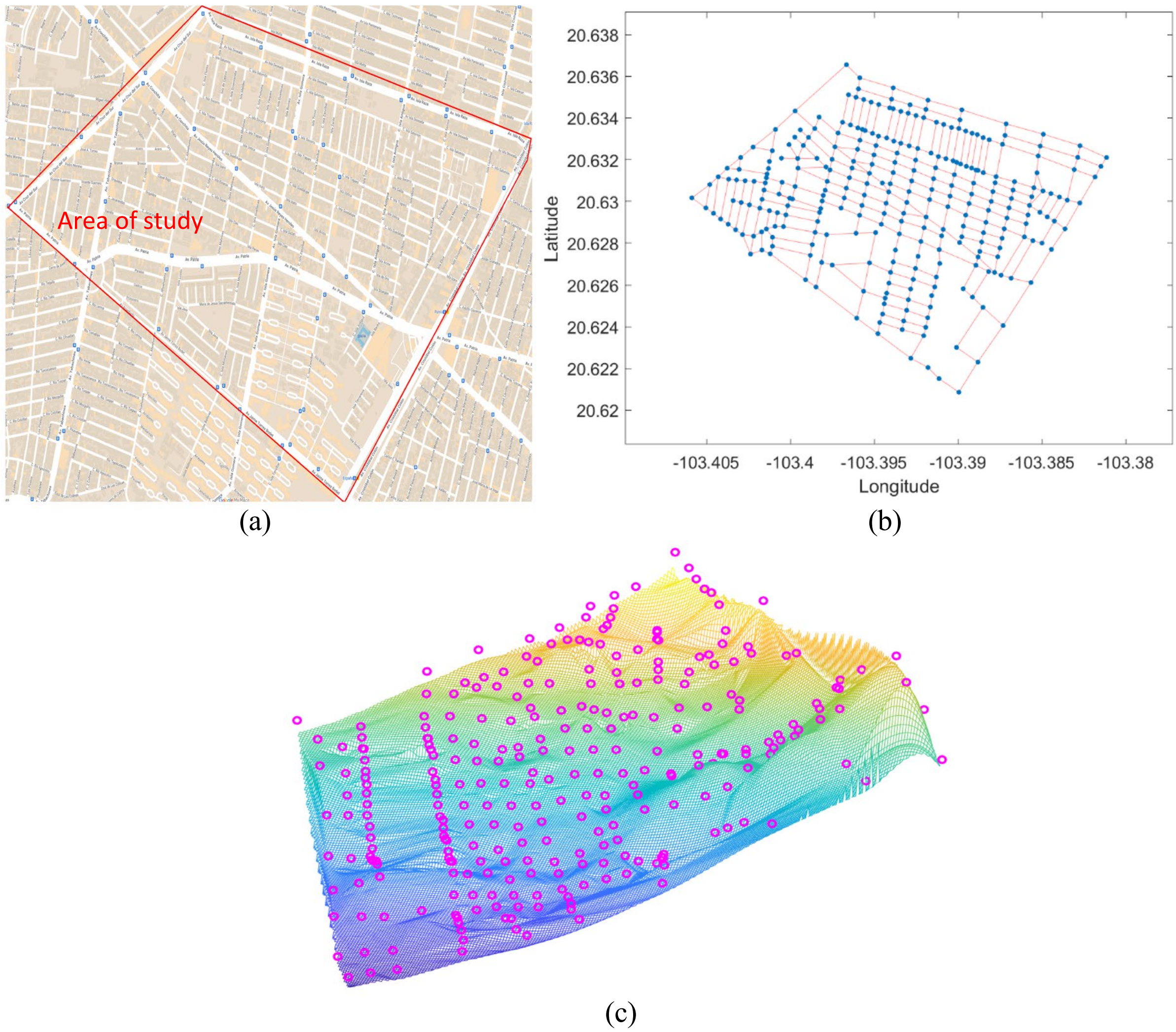


Fig. 6 Critical flood-prone area within Guadalajara, Jalisco, Mexico: **a** City map of the area to be analyzed, **b** resulting network, and **c** geographical position of each of the 285 nodes

5 Experimental results and validation analysis

In this section, the results and main characteristics of the proposed model are presented, and the proposed model is tested in various scenarios. Although the model is generic and can be applied to any region, area, or city, it was chosen to test it in a critical flood-prone area in Guadalajara, Jalisco, Mexico. Figure 6a highlights the specific area of study with a red perimeter, focusing on flood-prone sections within this boundary. Figure 6b shows the corresponding network with labeled latitude and longitude coordinates, and Fig. 6c shows the geographical position of each of the 285 nodes. For a better understanding of the graph, geographical data in the appendix was included. The objective here is to locate areas within the network where water tends to accumulate and potentially cause flooding. By observing the behavior of the network during a simulated rainfall event, the model can identify these critical points and propose the inclusion of storm drains as a solution. The experiment showcases how the model can assist in designing effective flood prevention strategies by suggesting specific storm drain locations. The second experiment focuses on analyzing the impact of the number and distribution of storm drains on flooding. The objective here is to investigate how different configurations of storm drains affect the overall flood risk within the network. By systematically reducing the number of storm drains and altering their distribution through random selection, the experiment examines the consequences of these changes on water accumulation and flood potential.

This experiment allows for a comprehensive exploration of the relationship between storm drain configuration and flood mitigation, providing insights into the importance of proper placement and quantity of storm drains. Both experiments aim to demonstrate the simplicity and flexibility of the proposed model. By dividing the experimental section into these two distinct tests, the study showcases how users can easily design specific scenarios and test them using the model.

5.1 First experiment: identifying critical points and suggesting storm drains to minimize flood risk.

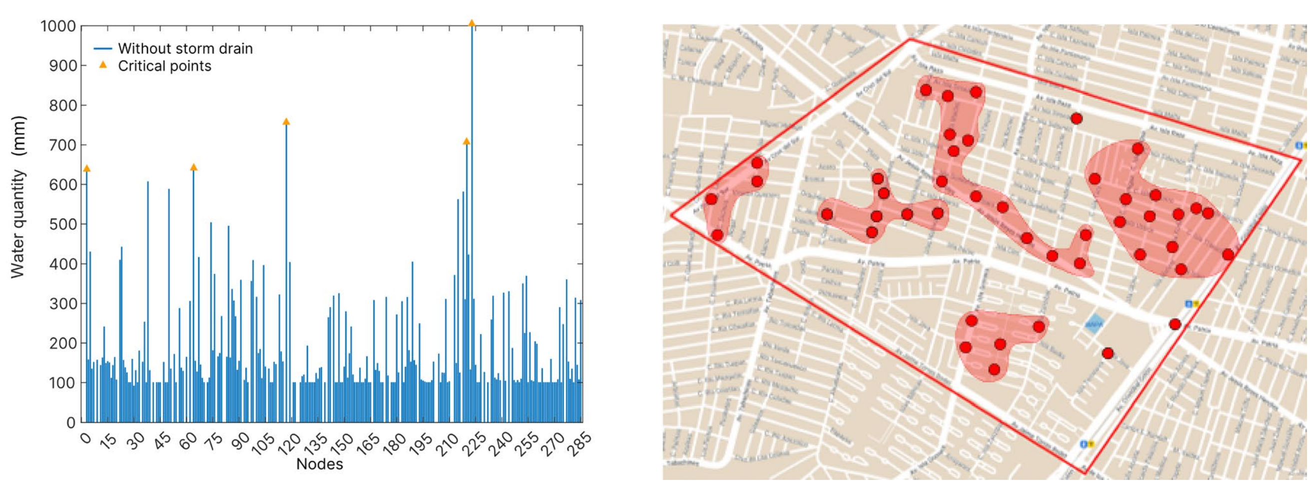
In the first experiment, all nodes in the network were assumed to start with the same amount of water, denoted as w_q, which was initialized as a random value within a uniform distribution $D[0{,}100]$ to simulate a standardized rainfall event. This assumption allows us to observe the baseline distribution of water across the network without introducing variability in the initial water volume. The diffusion rate for each node was also initialized within a uniform distribution $D[0{,}1]$, controlling the amount of water each node transfers to its neighbors. As each node interacts with its specific neighbors, the diffusion rate is influenced by factors such as elevation, number of adjacent nodes, number of sewers, quantity of garbage, infrastructure capacity, and the slope of connecting edges, allowing us to observe water accumulation in certain nodes based on the network's topological and geographical characteristics.

The first experiment used a diffusion model in complex networks to simulate flood situations in a specific context of urban hydrology. In this approach, the spread of rainwater is described by how it diffuses through a network of interconnected nodes, where each node represents a street intersection. The experiment aimed to identify the critical points in the network where water tends to accumulate and suggest possible solutions to avoid water concentrations. In the first step of the experiment, a complete network was observed, in order to observe how it behaved when a rainfall event occurred.

This helps identify the areas of the network that are prone to flooding and the critical points where water tends to accumulate. By analyzing the network behavior, it is possible to understand how water flows through the network and where it accumulates. Once the critical points are identified, the experiment suggests the inclusion of several storm drains to avoid water concentrations. Storm drains act as a release valve, allowing water to flow out of the network quickly, reducing the risk of flooding. By including these drains, the network performance can be improved, and the risk of flooding can be minimized.

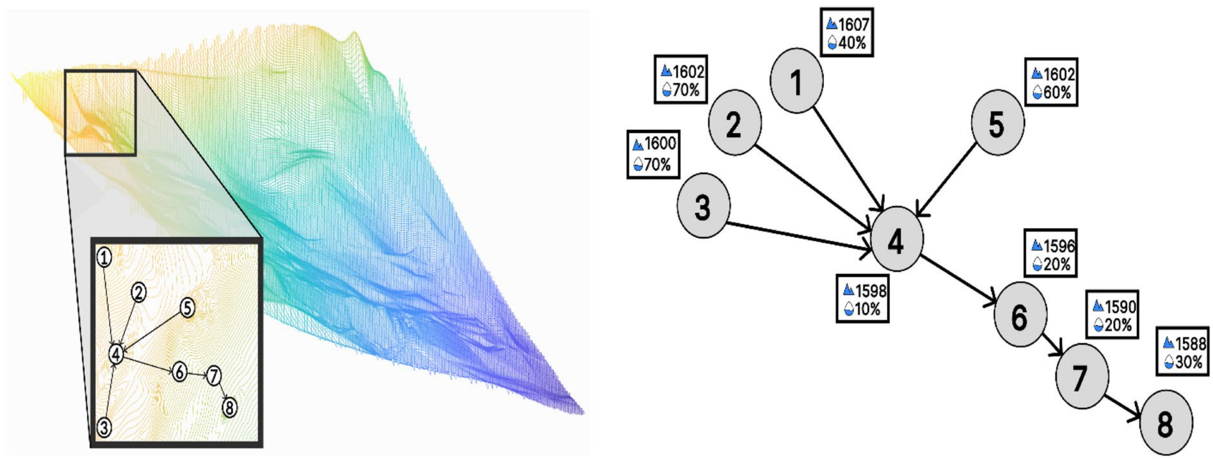
In the experiment, all nodes in the network were assumed to have the same amount of water, which is represented as wq. This means that each node initially had the same amount of water in the system. In addition, all nodes have the same diffusion rate, which is the rate at which water moves from one node to another. Under these conditions, the experiment was designed to emulate a normal storm in which all nodes receive the same amount of water. This allows for a fair comparison of the behavior of the complete network and helps identify critical points where water tends to accumulate. Figure 7a shows the results of the simulation, which shows the amount of water collected by each of the 285 nodes in the network.

The water levels are displayed in millimeters, representing the depth of accumulated water at each point in the network and providing a quantitative measure of water depth across different nodes. These results correspond to the 200th iteration of the model, where flood problems are already noticeable. It can be observed from the Figure that some nodes have higher water levels than others, which suggests that some areas are more prone to flooding. Figure 7b shows the positions of the nodes that present water levels higher



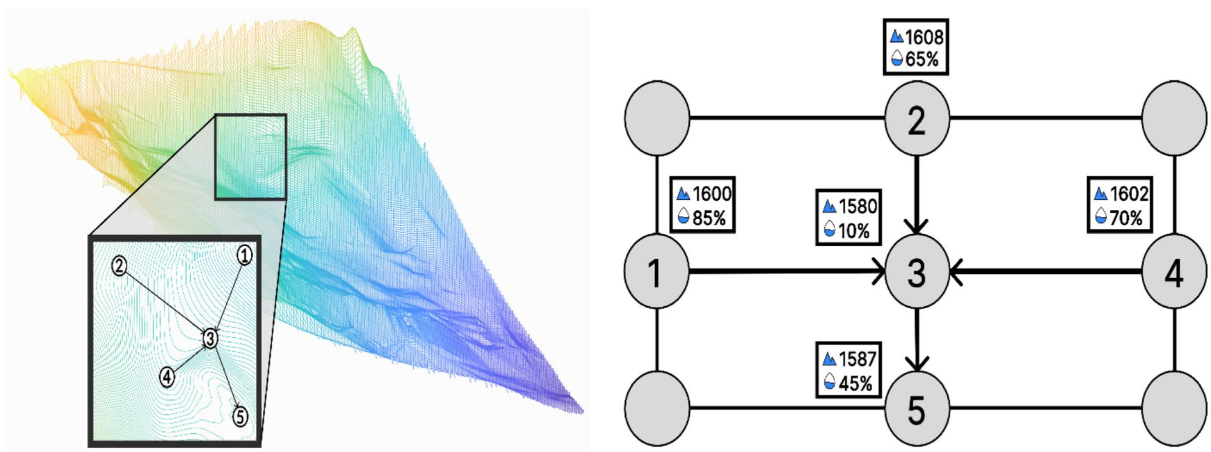
(a) (b)

Fig. 7 Results of the simulation: **a** water levels in each node and **b** positions of the nodes that present water levels higher than 300 mm of depth



(a) (b)

Fig. 8 The effects of position and topology on the accumulation of water for a node whose position and topology are prone to present high water levels. **a** Topology of the network in terms of landscape and **b** relative altitude of the nodes



(a) (b)

Fig. 9 Effects of position and topology on the accumulation of water for a node whose topology is prone to present high water levels. **a** Topology of the network in terms of landscape and **b** relative altitude of the nodes

than 300 mm. The nodes with high water levels are indicated in red and are grouped based on their proximity to each other. The grouping of nodes helps identify clusters of areas that are highly susceptible to flooding.

These results can be used to suggest measures to prevent or reduce the impact of flooding, such as the inclusion of storm drains or modification of the network topology in specific areas. High levels of water are influenced by two factors: the position and topology of nodes. Nodes located at lower positions, such as depressions or valleys, tend to receive more water than nodes located at higher positions.

This is because water naturally flows from higher to lower elevations; therefore, nodes located at lower positions are more likely to accumulate water. In addition to position, topology also plays a role in determining water accumulation in the nodes. Nodes that are topologically related to more neighbor nodes tend to collect more water from nodes than those that are related to fewer neighbor nodes.

This is because when water diffuses through the network, it tends to flow through nodes that are connected to more neighbors, leading to higher water accumulation in those nodes. Figures 7 and 8 show two different examples of flood situations that can arise in an urban hydrology context. In Figs. 8 and 9, the water accumulation shown represents initial states of the model, before the complete transfer of water toward lower-altitude nodes has occurred. As the simulation progresses through further iterations, water is gradually redistributed, moving from higher to lower nodes in accordance with the diffusion dynamics of the model. These initial states serve to illustrate the starting water distribution and highlight potential areas of accumulation, which will adjust over time as the model iterates and water flows

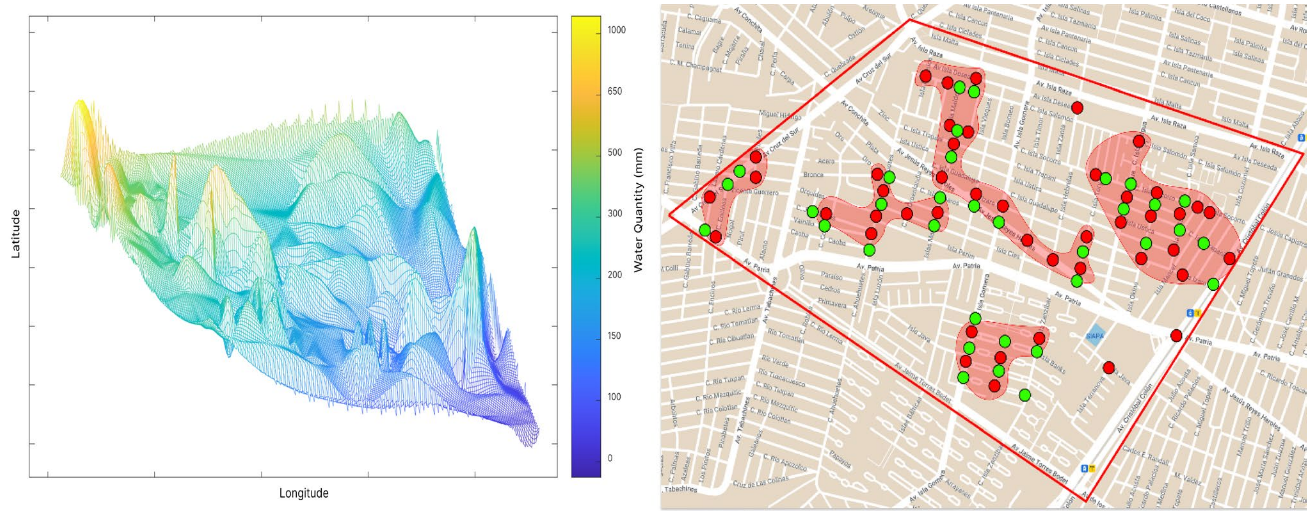


Fig. 10 Determination of suggested storm drains according to the water level collected by the nodes. **a** Water levels in the position of each node of the network and **b** suggested storm drains at n = 34 highest points

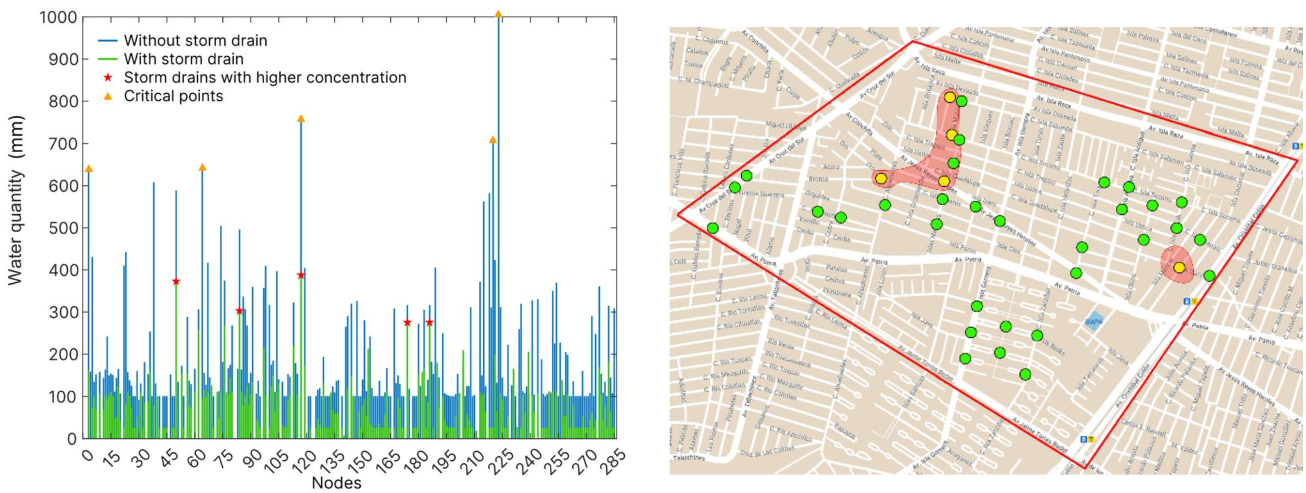


Fig. 11 Results of the simulation: **a** water levels in each node with and without storm drains, **b** position of the nodes; yellow nodes indicate water levels from 50 to 100 mm, and green nodes indicate points less than 50 mm

toward lower-altitude nodes. Figure 8 illustrates a situation in which high levels of water are produced by a combination of two factors: high differences in altitude and the transmission of water from multiple nodes to a single node.

In other words, water tends to accumulate at a specific location owing to both the topology and terrain of the area. Figure 8a shows the network topology in terms of the landscape. Figure 8b shows the relative altitudes of the nodes. On the other hand, Fig. 9 presents a situation where the nodes do not have a significant difference in altitude, but a single node collects a high level of water.

In this case, a high water level is caused by the transmission of water from multiple sources to a single node. Figure 8a shows the network topology. Figure 8b shows the relative altitudes of the nodes. Figure 9a shows the water levels at different positions in the complex network.

This image was produced by integrating the two different images. The first image shows an interpolation of the different water levels of the nodes based on their positions. This means that the water levels of the nodes are projected onto a continuous surface, allowing us to see how water accumulates and flows through the network.

The second image topologically represents the positions of the nodes in the network (Fig. 6c). By combining these two images, we can observe how the water levels correspond to the physical characteristics of the landscape. Once the water levels in each node are obtained through interpolation, we can use this information to locate the most high-risk points and suggest the location of storm drains to avoid flood conditions.

In Fig. 10b, red points indicate the nodes with the highest water levels, where storm drains are suggested to mitigate potential flooding, while green points represent the suggested locations of the storm drains, which are based on the highest water levels (n=34). These storm drains are strategically placed to reduce the impact of flood conditions caused by incorrect positions and configurations of nodes in the network. After considering the effects of storm drains, the simulation was executed, and the results are shown in Fig. 11a. The green bars represent the simulation results with storm drains, while the blue bars show the results without considering the effects of storm drains.

A comparison between both scenarios demonstrates the effectiveness of the proposed solution in reducing the impact of floods. Figure 11b shows the effects of storm drains on the simulation in terms of the water levels in the nodes. The yellow nodes indicate the locations where the water level was significantly reduced (less than 100 mm) as a result of the presence of storm drains, while the green nodes represent water levels of less than 50 mm.

These results demonstrate the effectiveness of the proposed approach for mitigating the impact of floods on the network. Figure 10b shows the positions of the nodes that presented a level of water greater than 300 mm in depth after considering the placement of storm drains. Comparing this Figure to Fig. 6b, which shows the same information without considering storm drains, it is clear that the effects of storm drains have considerably reduced the flood impact. Figure 6b shows several clusters of nodes with high water levels, indicating areas that are particularly vulnerable to flooding. In contrast, in Fig. 11b, these clusters have been greatly diluted, indicating that the storm drains effectively redirected the excess water and prevented it from accumulating in certain areas.

5.2 Second experiment: analyzing the impact of the number and distribution of storm drains on flooding

In the second experiment, the initial water quantity and diffusion rate for each node were assigned following the same procedure as in the first experiment, with the initial water quantity w_q set as a random value within $D[0,100]$ and the diffusion rate initialized within $D[0,1]$. These values, influenced by factors such as elevation, neighboring nodes, and slope, provided a baseline for testing the impact of storm drain configurations on water distribution and flood risk across the network.

In this experiment, we analyze the process of locating storm drains using our model to mitigate the effects of flooding. We emphasize that the effectiveness of the storm drains depends on their number, as well as their position and distribution within the network.

To investigate this, we conducted several tests to assess the impact of different numbers and distributions of storm drains on flood reduction. To simulate a variety of scenarios and prevent overfitting, the storm drains in each test were distributed randomly. This approach allows the model to generalize more effectively, providing flexibility and robustness by exploring a range of potential outcomes under varying configurations. Although some storm drains, such as drain T, may occupy strategically advantageous positions, a more refined analysis involving optimized distributions could further enhance the model's predictive capabilities in future studies. The objective of the experiment is to demonstrate that individuals without expertise in urban hydrology can effectively utilize the proposed model to determine the appropriate configuration of storm drain elements for a specific network. This objective is motivated by the need to empower non-experts to make informed decisions regarding flood prevention and water management in urban areas.

Initially, we considered 34 drain storm elements strategically placed at locations where the nodes had the highest water levels. These locations were identified based on the results obtained from the previous experiment using our model. By targeting areas with significant water accumulation, we aimed to optimize the efficiency of the storm drains in reducing flooding. To evaluate the influence of the number of storm drains on flood mitigation, we employed a random selection process.

This process progressively reduced the number of drain storm elements, resulting in final configurations with 80%, 75%, 50%, and 15% of the initial number of elements, respectively. This allowed us to observe the impact of decreasing the number of storm drains on flood reduction capabilities. By conducting these tests, we were able to assess the effectiveness of different configurations of storm drains in mitigating flood risks.

We analyzed the resulting flood scenarios, comparing them to the original simulation without storm drains. Through careful observation and data analysis, we measured the extent to which each configuration of storm drains contributed to reducing the water levels in the network.

In the first test of our experiment, we aimed to assess the impact of reducing the number of storm drains to 80% of their original quantity. In this particular test, the selection of storm drains was performed randomly without considering their distribution within the network. The configuration of the selected storm drains can be observed in Fig. 12a, where the green points represent the storm drain locations.

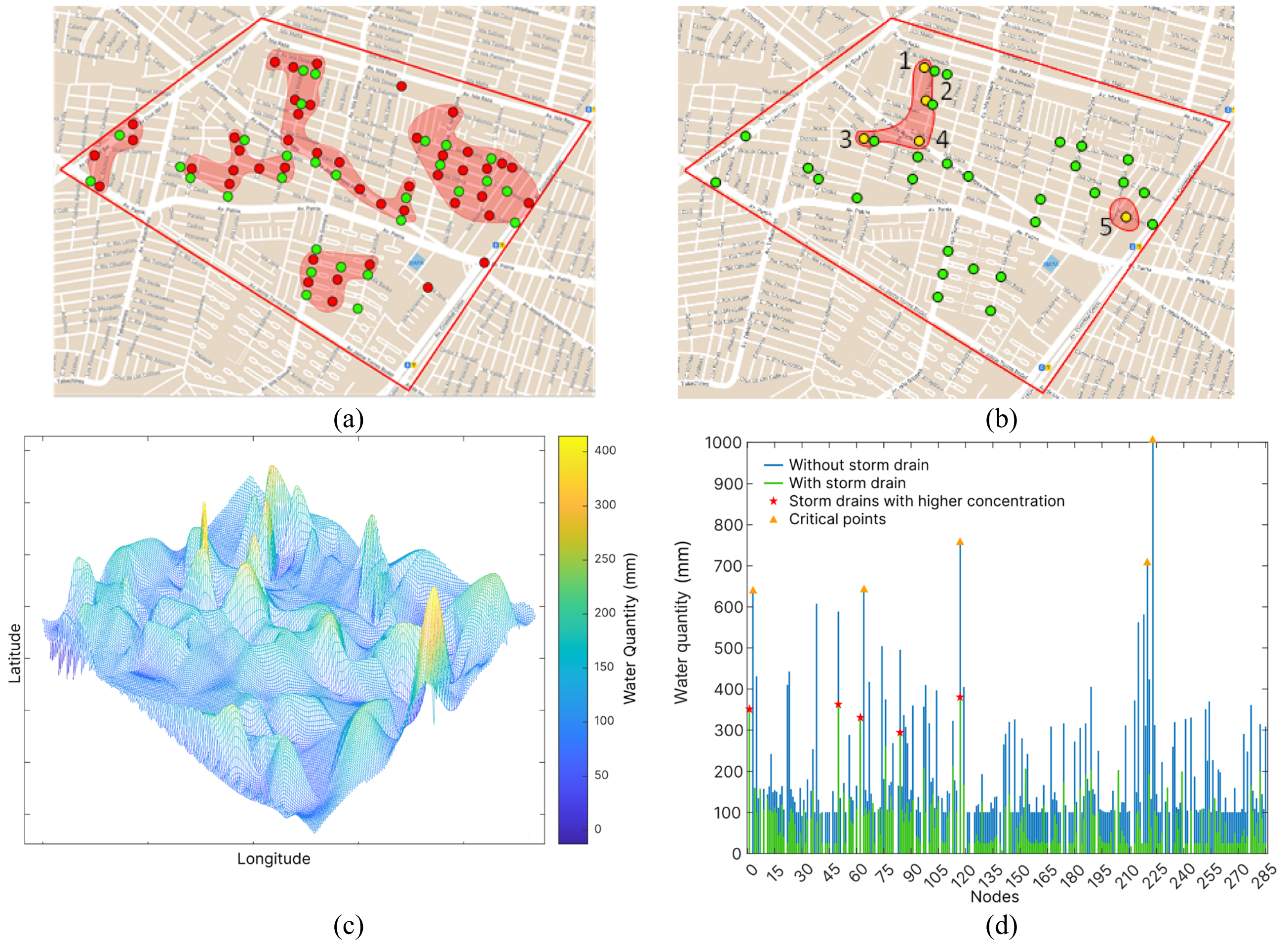


Fig. 12 Results for the case of considering 80% of the drain storm elements

To evaluate the effectiveness of this configuration, we conducted simulations using our model. The results of the simulation are presented in Fig. 12b.

From Fig. 12, it is evident that only five nodes exhibit a higher concentration of water, which are highlighted in yellow color. The analysis of the simulation results provides valuable insights into the spatial distribution of water accumulation within the network. In this test, one notable observation is that a majority of the yellow nodes (elements with a higher concentration of water) are concentrated in a specific region. Importantly, none of the nodes reached a flood condition, indicated by the absence of red points in the Figure.

Furthermore, Fig. 12c depicts a surface representation illustrating the distribution of water across the network. The yellow points observed in Fig. 12b are clearly visible in this Figure, indicating the nodes with the highest water levels. To further analyze the impact of the reduced number of storm drains, Fig. 12d provides the water content of each node considering 80% of the total storm drains (represented by green values).

Additionally, the blue values in the same Figure represent the scenario where no storm drains are considered, serving as a reference point for comparison. After conducting the simulations, the results clearly indicate that even with the removal of 20% of the storm drains, the possibility of flooding is not high.

In the second test of our experiment, we further reduced the number of storm drains to 75% of the original quantity. Similar to the previous test, the selection of the storm drains was done randomly, without considering their distribution within the network. Figure 13a illustrates the configuration of the selected storm drains, represented by green points. Under these experimental conditions, the result of the simulation is presented in Fig. 13b.

One notable difference from the previous test, where 80% of the storm drains were considered, is that a majority of the nodes with a higher concentration of water are now distributed throughout the entire network. This indicates that the reduction in the number of storm drains, without optimizing their distribution, has led to a more dispersed water accumulation pattern.

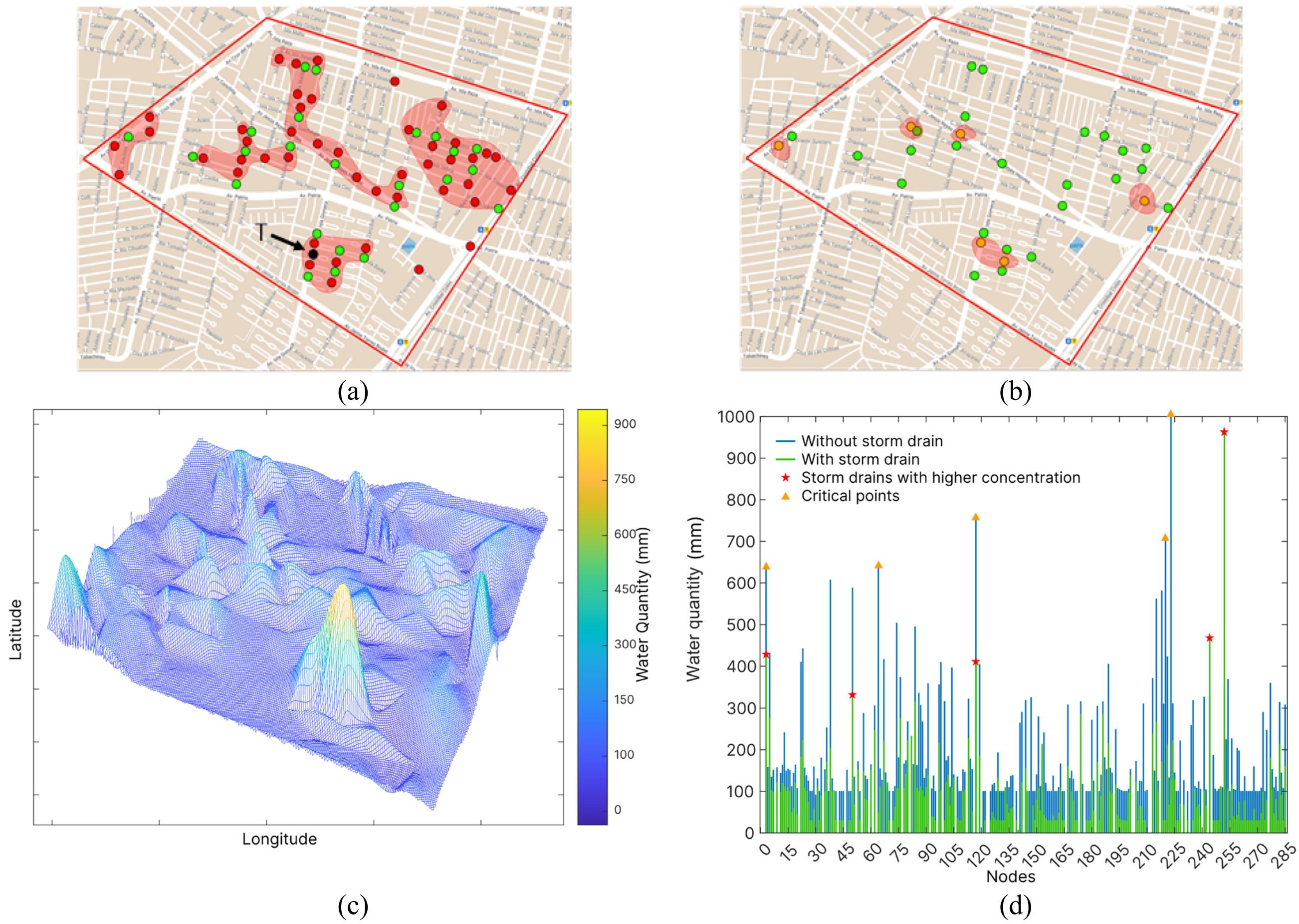


Fig. 13 Results for the case of considering 80% of the drain storm elements

Figure 13c provides a surface visualization of the water distribution across the network. Here, it becomes evident that without an optimized configuration of storm drains, a distinct peak of water concentration emerges.

This peak represents an area of significant water accumulation and suggests that the remaining number of storm drains does not provide an optimal solution to mitigate the flooding problem. To better understand the impact of reducing the number of storm drains, Fig. 13d displays the water content of each node. The green values represent the scenario where 75% of the total storm drains are in place, while the blue values serve as a reference where no storm drains are considered.

Additionally, Fig. 13a highlights a black point (labeled with T) symbolizing the location of a storm drain that has been deleted for this particular test. It is important to note that this storm drain, which has been removed, plays a crucial role in preventing flooding, as will be demonstrated in subsequent tests. The peak observed in Fig. 13c, which represents a concentrated area of water accumulation, can be attributed to the elimination of a specific storm drain, denoted as T. This storm drain plays a crucial role in preventing water accumulation due to its strategic position and the number of neighboring nodes that transmit water to it.

In the third test, the number of storm drains is reduced to 50% of its original number, and their positions are selected randomly without considering their distribution. The configuration of the selected storm drains is depicted in Fig. 14a, where the green points represent the storm drain locations. Under these experimental conditions, the simulation results are presented in Fig. 14b. In this test, a significant reduction in the number of storm drains has been implemented. However, despite this reduction, it is evident that although more nodes exhibit water accumulation, the water levels in these nodes are not high. This indicates that the network system does not experience flooding even with fewer storm drains. Figure 14c displays the water distribution surface, which illustrates that, unlike the test with 75% of the storm drains, the previously excluded storm drain (denoted as black) has been reintroduced and considered valid in the simulation. The presence of this storm drain plays a critical role in preventing high water levels within the network. It is apparent that no significant peaks of water concentration are

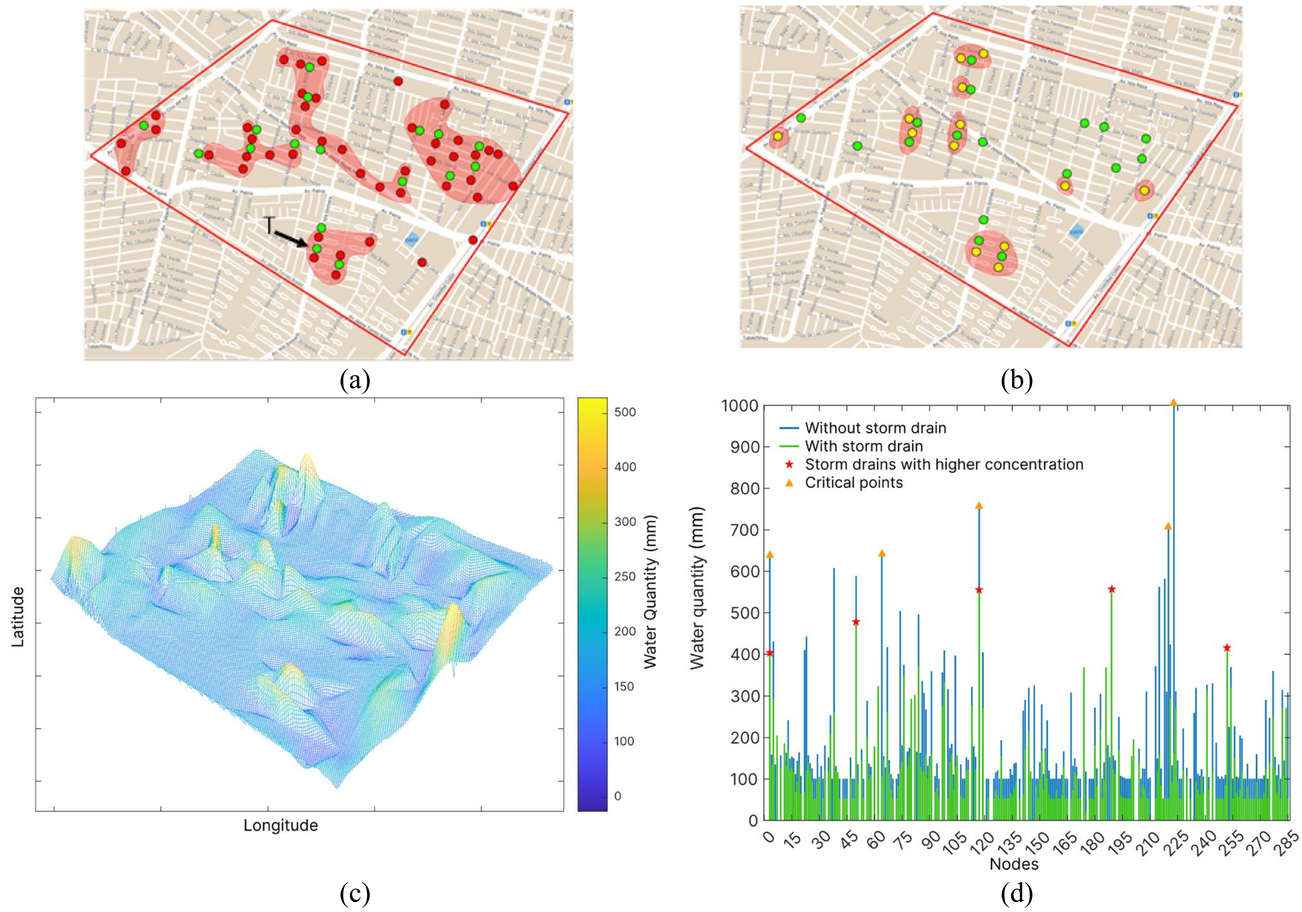


Fig. 14 Results for the case of considering 50% of the drain storm elements

observed in Fig. 14c, unlike the peak observed in Fig. 13c, resulting from the absence of storm drain T in the 75% test.

The effect of the reintroduced storm drain is highly influential in maintaining water levels within acceptable limits throughout the network. It acts as a crucial element in preventing water accumulation and potential flooding. The strategic positioning and effectiveness of this storm drain ensure that water is efficiently drained and distributed, mitigating the risk of excessive water accumulation in specific areas. Figure 14d provides detailed information on the water content of each node, considering the presence of 50% of the total storm drains (green values). The blue values represent the scenario where no storm drain is present and are provided as a reference. These findings highlight the significance of the reintroduced storm drain in the prevention of flooding. Its inclusion in the network system ensures that water levels remain at manageable levels, even with a reduced number of storm drains.

The absence of pronounced peaks in Fig. 14c further confirms the effectiveness of the storm drain in maintaining a relatively uniform distribution of water throughout the network.

In the fourth test, the number of storm drains is reduced to 15% of its original number, representing the minimal number of storm drains explored in this experiment. The selected storm drain locations are determined randomly without considering their distribution. The configuration of the selected storm drains is depicted in Fig. 15a, where the green points represent the storm drain positions. Under these experimental conditions, the simulation results are presented in Fig. 15b. In this test, the number of storm drains has been drastically reduced to the minimum possible. However, despite this significant reduction, it is evident that although certain nodes exhibit water accumulation, the water levels in these nodes remain relatively low. This suggests that the system does not experience high levels of flooding even with a significantly reduced number of storm drains. Figure 15c shows the water distribution surface, which indicates the spatial distribution of water throughout the network. Similar to the previous tests, the black storm drain (T) is included as a valid element in our simulation, playing a crucial role in preventing high water levels. Notably, no remarkable peaks of water concentration are

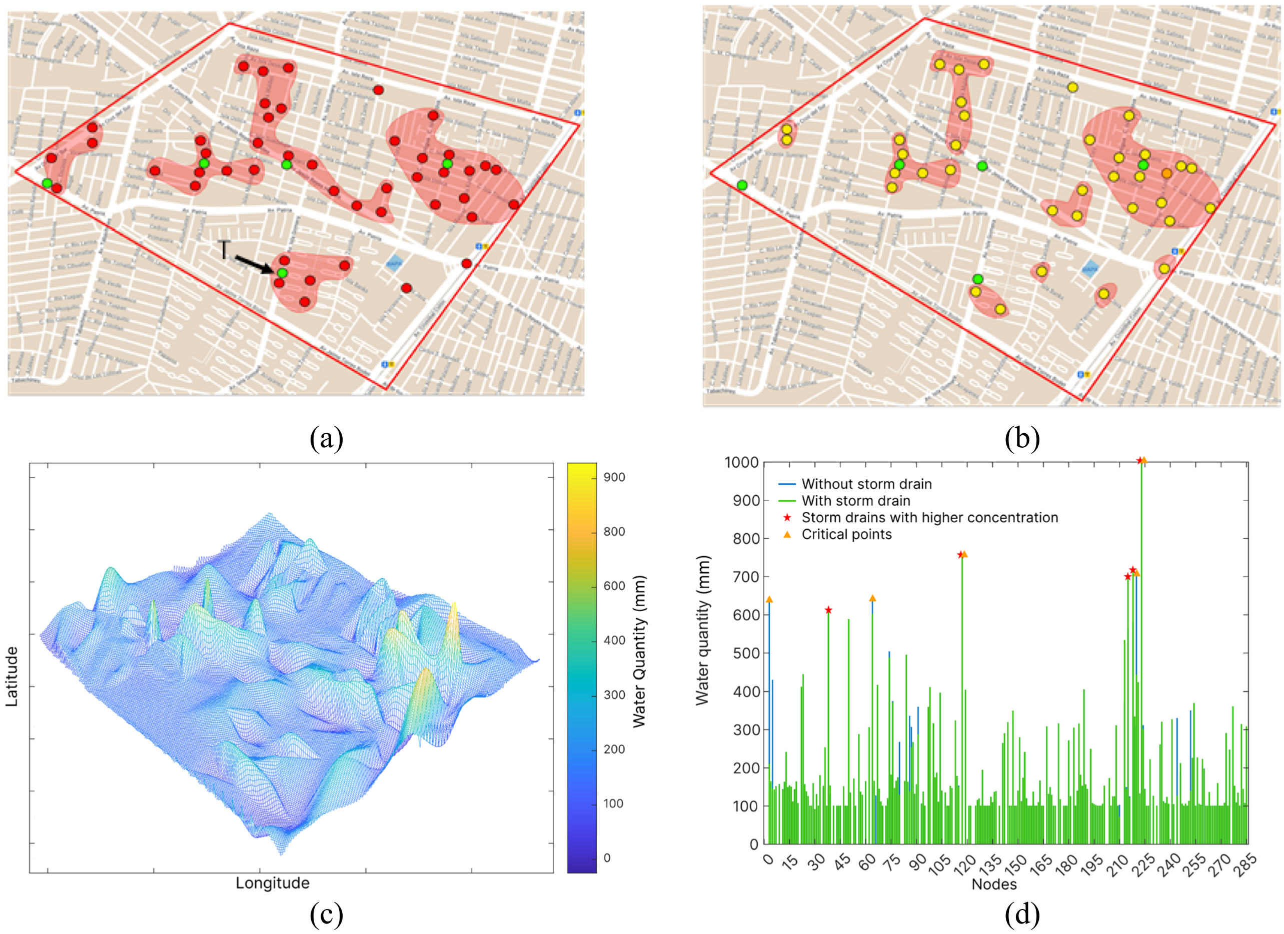


Fig. 15 Results for the case of considering 15% of the drain storm elements

observed in Fig. 15c, indicating that storm drain T effectively mitigates excessive water accumulation.

Figure 15d provides detailed information on the water content of each node, considering the presence of only 15% of the total storm drains (green values). The blue values represent the scenario where no storm drain is present, serving as a reference. According to Fig. 15d, the water levels in the simulated system align with the low water levels observed in the system without any storm drains. However, it is important to note that the system with 15% of the storm drains exhibits a filtering effect on high-water levels.

These findings highlight the critical role of the storm drain network, even when reduced to a minimal number of elements. The inclusion of strategically positioned storm drains ensures that water levels throughout the network are maintained at manageable levels, effectively preventing flooding. The absence of significant peaks in Fig. 15c further emphasizes the importance of the storm drain in maintaining a balanced water distribution.

6 Validation of the model

Although the primary motivation for using an agent-based model (ABM) is to visualize how emergent, unscheduled patterns arise from agents' interactions with each other and various environmental elements, our paper emphasizes this capability by tracking flood patterns in interconnected agent chains. Specifically, we have highlighted how certain agents—representing locations or structures in an urban setting—become flooded due to the way they interact with incoming water flows from neighboring agents. By allowing each agent to respond to and transmit water flow based on localized conditions, the model enables dynamic, step-by-step observation of how flood zones propagate. These emergent patterns provide valuable insights into potential conflict zones within the network, where water accumulation poses the highest risk. Identifying these chains of flood-prone agents allows us to pinpoint critical areas requiring intervention, making the ABM an effective tool for early detection and strategic flood management.

A significant advancement in our approach was validating whether the water flows produced by certain agents due to interactions with neighboring elements aligned with measurements recorded by the civil protection authorities in Guadalajara, Mexico. While a direct comparison was not possible for all agents, given that flow and flood sensors are installed only at five of the most critical points for cost-efficiency and monitoring ease, we focused on nodes a_3, a_{64}, a_{117}, a_{220}, and a_{223}—identified as the highest-risk areas in our analysis. Upon comparing the simulated flow values of these specific agents to the data captured by the sensors, we observed a discrepancy of less than 10%. For a system as complex as urban hydrology, this margin of error is sufficiently low to affirm the model's accuracy and robustness. This close alignment between the simulated and actual flows underscores the model's effectiveness in realistically capturing water dynamics within urban flood-prone zones, reinforcing its value as a predictive and planning tool.

7 Conclusion

In this paper, the diffusion model in complex networks is employed to simulate flood scenarios in the context of urban hydrology. The model is adapted to describe how rainfall water spreads through interconnected nodes within the network, representing various areas of a city such as buildings, roads, and water bodies.

Each node is assigned specific properties, such as elevation, surface roughness, and permeability, which influence the flow of water. The edges connecting the nodes represent the pathways for water movement. The model takes into account factors such as rainfall intensity, network topology, properties of the nodes, and existing flood mitigation measures.

To evaluate the performance of the proposed model, two experiments were conducted. The first experiment aimed to identify critical points in the network and suggest optimal storm drain placements to minimize flood risk. This involved analyzing the impact of different storm drain configurations on flood mitigation. The second experiment focused on studying the influence of the number and distribution of storm drains on flooding within the network. By varying these factors, the researchers could assess their effects on flood levels and extent.

To validate the model, a case study was conducted in a flood-prone area of Guadalajara, Mexico. The model was applied to simulate various flooding scenarios, considering different rainfall patterns and storm drain configurations. The experimental results demonstrate the flexibility and simplicity of the model, allowing for insightful conclusions to be drawn regarding flood risk management strategies. The findings contribute to the understanding of flood dynamics in urban areas and can aid in the development of effective flood mitigation measures.

Author contributions We are pleased to submit our manuscript titled "Simulating Flood Situations in Urban Hydrology Using a Diffusion Model in Complex Networks" for consideration for publication in your esteemed journal. The work presented in this manuscript represents a collaborative effort among five authors, each of whom made significant contributions to the research. Erik Cuevas played a critical role in the writing process and the design of the proposed method. Miguel Toski and Hector Escobar was responsible for conducting the experiments and designing the approach. Bernardo Morales and Marco Peréz contributed to the discussions and the writing process. Overall, the collaboration among the five authors was key to the success of this project. We are confident that the work presented in this manuscript will be of great interest to your readers, and we look forward to the opportunity to share our findings with the scientific community. Thank you for considering our submission. Sincerely, Miguel Toski, Erik Cuevas, Hector Escobar, Bernardo Morales, and Marco Peréz.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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